

Modernizing Digital Banking Operations at a Retail Bank

Final Project | Systems Analysis & Design

ABC Bank | Digital Transformation Initiative

Executive Summary

THE PROBLEM

- Legacy systems slow onboarding to 5+ days
- Manual processes cause errors and delays
- Poor system integration across channels

KEY INSIGHTS

- Cloud-native microservices approach
- API-first integration strategy
- Containerized deployment model

RECOMMENDED ACTIONS

- 1 Migrate to IBM Cloud platform
- 2 Implement CI/CD pipelines
- 3 Deploy event-driven architecture
- 4 Modernize database layer
- 5 Establish real-time monitoring

Introduction

THE OPPORTUNITY

- Modernize ABC Bank's customer-facing and back-office systems
- Replace legacy monolithic architecture with cloud-native services
- Improve onboarding, transactions, and loan processing workflows

APPROACH

- UML and BPMN process modeling for requirements capture
- Data flow and architecture analysis for system design
- Agile methodology with phased delivery across three phases

KEY QUESTIONS AND HYPOTHESES

- How to reduce customer onboarding time by 60% through automation?
- How to ensure 99.9% system uptime with cloud-native architecture?
- How to meet PCI-DSS and KYC regulatory compliance while modernizing?
- Can event-driven architecture handle real-time fraud detection at scale?
- Will containerized microservices reduce deployment cycles from monthly to weekly?
- How to manage change resistance and ensure staff adoption of new systems?

Objectives

60%

Reduce onboarding
from 5 to 2 days

99.9%

System availability
target uptime

80%

Transaction
automation rate

Real-time

Fraud detection
implementation

PCI-DSS

Full KYC and
security compliance

40%

Customer satisfaction
score improvement

Project Charter

Element	Details
Project Title	Digital Banking Modernization Initiative
Date	January 15, 2025
Project Description	End-to-end modernization of ABC Bank's core banking operations from legacy monolithic systems to cloud-native microservices
Objectives	Reduce onboarding from 5 to 2 days; achieve 99.9% uptime; automate 80% of transactions; implement real-time fraud detection
Success Criteria	Onboarding < 2 days; uptime >= 99.9%; 80% transaction automation; PCI-DSS audit pass; zero critical security breaches
High-Level Requirements	Cloud migration; microservices architecture; CI/CD pipeline; real-time monitoring; automated KYC; event-driven processing
Key Stakeholders/Roles	CTO (Sponsor); IT Director (Tech Lead); Branch Manager (End User); Compliance Officer (Advisor); CS Lead (End User)
Scope	Core banking, customer onboarding, loan processing, payment transactions, fraud detection
Deliverables	Cloud platform on IBM Cloud; microservices suite; CI/CD pipeline; monitoring dashboard; security framework
Assumptions	Existing infrastructure supports hybrid cloud; staff available for training; regulatory approvals obtained in parallel
High-Level Risks	Legacy migration complexity; data loss during migration; staff resistance; cybersecurity threats; regulatory non-compliance
Milestones	Requirements (Wk 2); Architecture (Wk 4); Infrastructure (Wk 8); Dev complete (Wk 20); UAT (Wk 28); Go-live (Wk 32)
Budget	\$2.4M total: Infrastructure \$800K; Development \$1M; Testing \$300K; Training \$150K; Contingency \$150K

Stakeholder Register

Stakeholder	Role	Attitude	Interest	Impact	Influence	Comm. Pref.	Location	Success Criteria	Hours
CTO	Sponsor	Supportive	High	High	High	Weekly brief	HQ Floor 10	On-time delivery, budget compliance	9-5
IT Director	Tech Lead	Neutral	High	High	High	Bi-weekly	HQ Floor 8	System stability, zero downtime	9-6
Branch Mgr	End User	Resistant	Medium	Medium	Medium	Monthly	Branch Office	Workflow efficiency gains	8-5
Compliance	Advisor	Supportive	High	High	Medium	Weekly	HQ Floor 9	PCI-DSS pass, KYC automated	9-5
CS Lead	End User	Supportive	Medium	Medium	Low	Training	Call Center	Reduced customer complaints	Shift

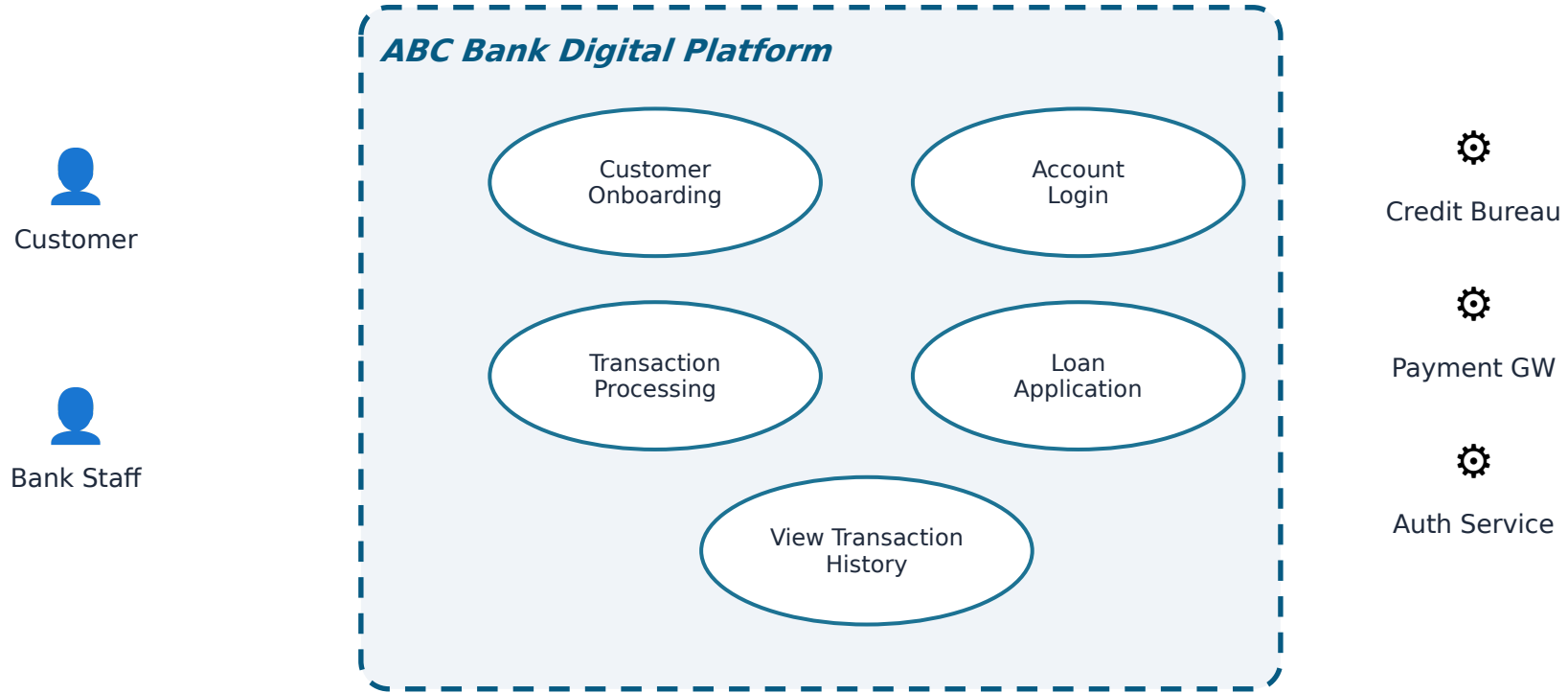
Stakeholder Engagement Plan

Stakeholder	Current State	Desired State	Strategy
CTO	Leading	Leading	Regular progress demos and KPI dashboards
IT Director	Neutral	Supportive	Involve in architecture decisions early
Branch Manager	Resistant	Neutral	Show workflow efficiency benefits early
Compliance Officer	Supportive	Leading	Co-design compliance checks and audits
CS Lead	Supportive	Supportive	Provide training and feedback loops

Business Requirements Document

Section	Details
Scope	End-to-end digital banking platform covering onboarding, transactions, and loans
Objectives	Modernize onboarding, payments, and loan processing workflows
Assumptions	Existing infrastructure supports hybrid cloud; staff available for training
Success Metrics	Onboarding < 2 days 99.9% uptime 80% automation Zero critical breaches

Use Case Diagram



Relationships: <<include>> Authentication / <<include>> Credit Check / <<extend>> Fraud Alert

Requirements Traceability Matrix

Req ID	Type	Description	Source	Biz Objective	Acceptance Criteria	Linked Req	Owner	Priority	Status	Verification	Risk
REQ-001	Func	Online account opening	CTO	Reduce onboarding 60%	Complete < 10 min	REQ-003, REQ-004	IT Dir.	High	In Progress	UAT	R-003
REQ-002	Func	Real-time transactions	Br. Mgr	80% automation	Process < 2 sec	REQ-003	Dev Lead	High	In Progress	Perf Test	R-001
REQ-003	NFR	99.9% system uptime	CTO	System availability	< 8.7 hrs down/yr	REQ-001, REQ-002	IT Dir.	Critical	Planned	Monitoring	R-002
REQ-004	Func	Automated KYC	Comp.	PCI-DSS compliance	Verify < 5 min	REQ-001, REQ-005	Comp Lead	High	In Progress	Int. Test	R-004
REQ-005	NFR	PCI-DSS compliance	Comp.	Security compliance	Pass annual audit	REQ-004	Sec Lead	Critical	In Progress	Sec Audit	R-005

Work Breakdown Structure

1.0 Digital Banking Modernization

1.1 Planning

1.1.1
Requirements
gathering

1.1.2
Stakeholder
analysis

1.1.3 Risk
assessment

1.1.4
Architecture
design

1.2 Execution

1.2.1
Infrastructure
setup

1.2.2
Microservices
development

1.2.3 Data
migration

1.2.4
Integration
testing

1.3 Closure

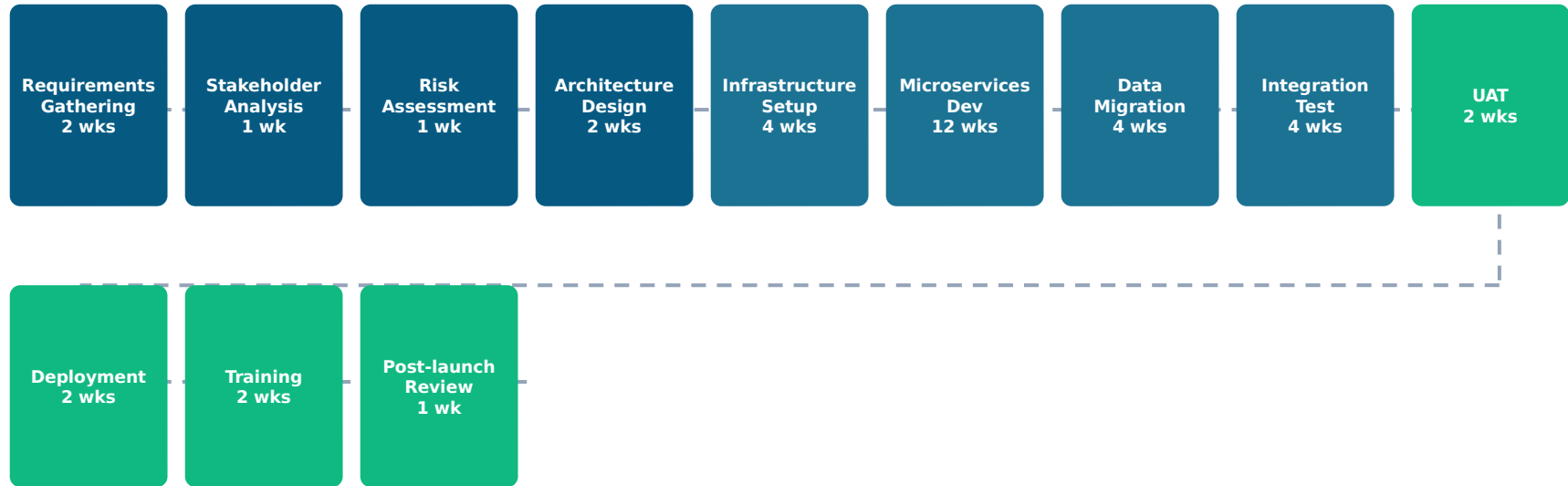
1.3.1 UAT

1.3.2
Deployment

1.3.3
Training

1.3.4 Post-
launch
review

Network Diagram



Critical Path: Requirements → Stakeholder Analysis → Risk Assessment → Architecture → Infrastructure → Microservices → Data Migration → Integration Test → UAT → Deployment → Training → Post-launch Review

Total Duration: ~36 weeks | Planning: 6 wks | Execution: 24 wks | Closure: 7 wks

SWOT Analysis

STRENGTHS

- Strong executive sponsorship from CTO
- Existing large customer base of 2M users
- Skilled IT team with Java and cloud experience
- Clear regulatory framework and compliance history

WEAKNESSES

- Heavy legacy system dependencies and technical debt
- Limited cloud-native and Kubernetes expertise
- Change resistance from branch operations staff
- Fragmented data across siloed legacy databases

OPPORTUNITIES

- Cloud-native scalability for peak transaction loads
- AI-driven fraud detection and risk scoring
- Open banking APIs for third-party integration
- Market differentiation through digital-first experience

THREATS

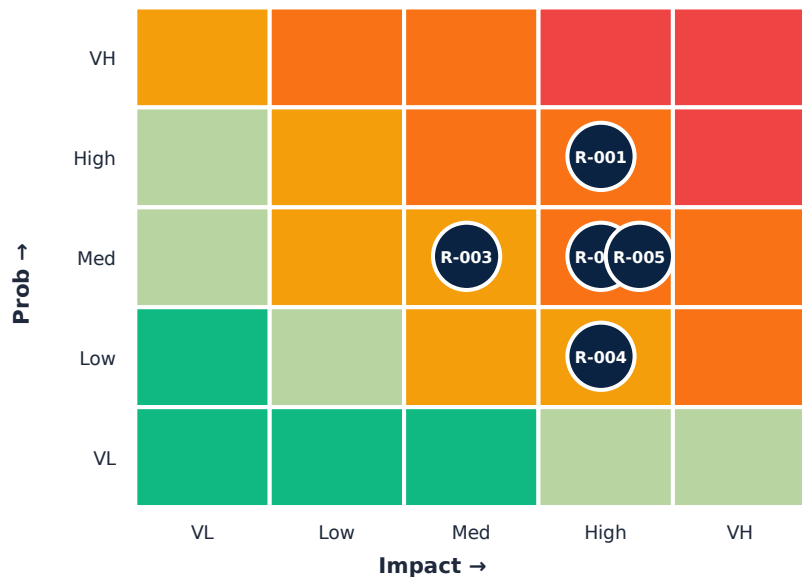
- Cybersecurity risks during migration window
- Regulatory changes mid-implementation
- Competitor banks launching digital platforms first
- Data migration failures causing service outages

Risk Register

ID	Cause	Event	Impact	Owner	Cat.	Prob	Impact	Rating	Score	Trigger	Response
R-001	Legacy system complexity	Migration failure during cutover	Service downtime	IT Dir.	Tech	High	High	Critical	9	Failed test migration	Rollback plan
R-002	Single point of failure	Production system outage	Revenue loss	IT Dir.	Tech	Med	High	High	6	Server health alerts	Auto-failover
R-003	Manual onboarding process	Customer onboarding delayed	Customer churn	Br. Mgr	Ops	Med	Med	Med	4	SLA breach on onboarding	Process automation
R-004	Regulatory gaps in new system	Compliance audit failure	Financial penalties	Comp.	Comp	Low	High	Med	3	Pre-audit findings	Compliance review
R-005	External cyber threats	Data breach incident	Reputation damage	Sec Lead	Sec	Med	High	High	6	Intrusion detection alert	Incident response

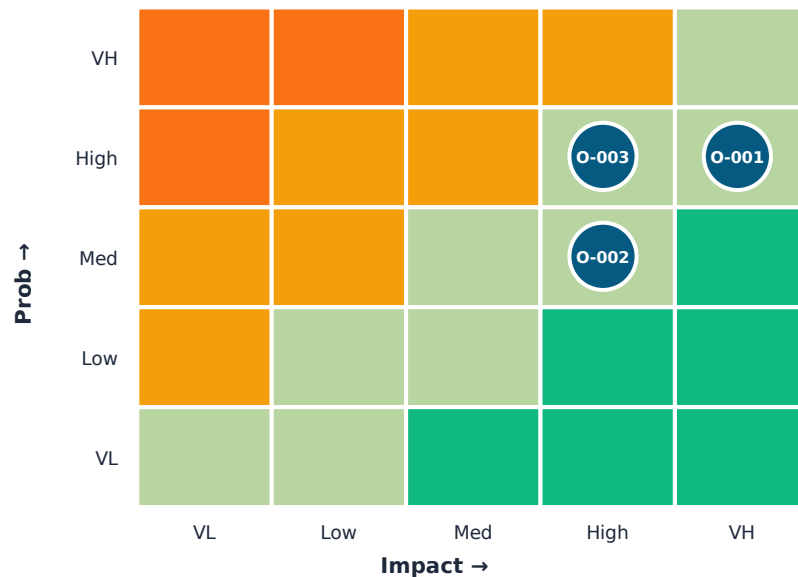
Risk Matrix: Threat and Opportunity

THREAT MATRIX



Threats: R-001 Migration failure | R-002 System outage | R-003 Onboarding delay | R-004 Compliance failure | R-005 Data breach

OPPORTUNITY MATRIX



Opportunities: O-001 Cloud-native scalability | O-002 AI-driven fraud detection | O-003 Open banking APIs

Core Banking Process Flowchart

CUSTOMER ONBOARDING



TRANSACTION PROCESSING



LOAN APPLICATION

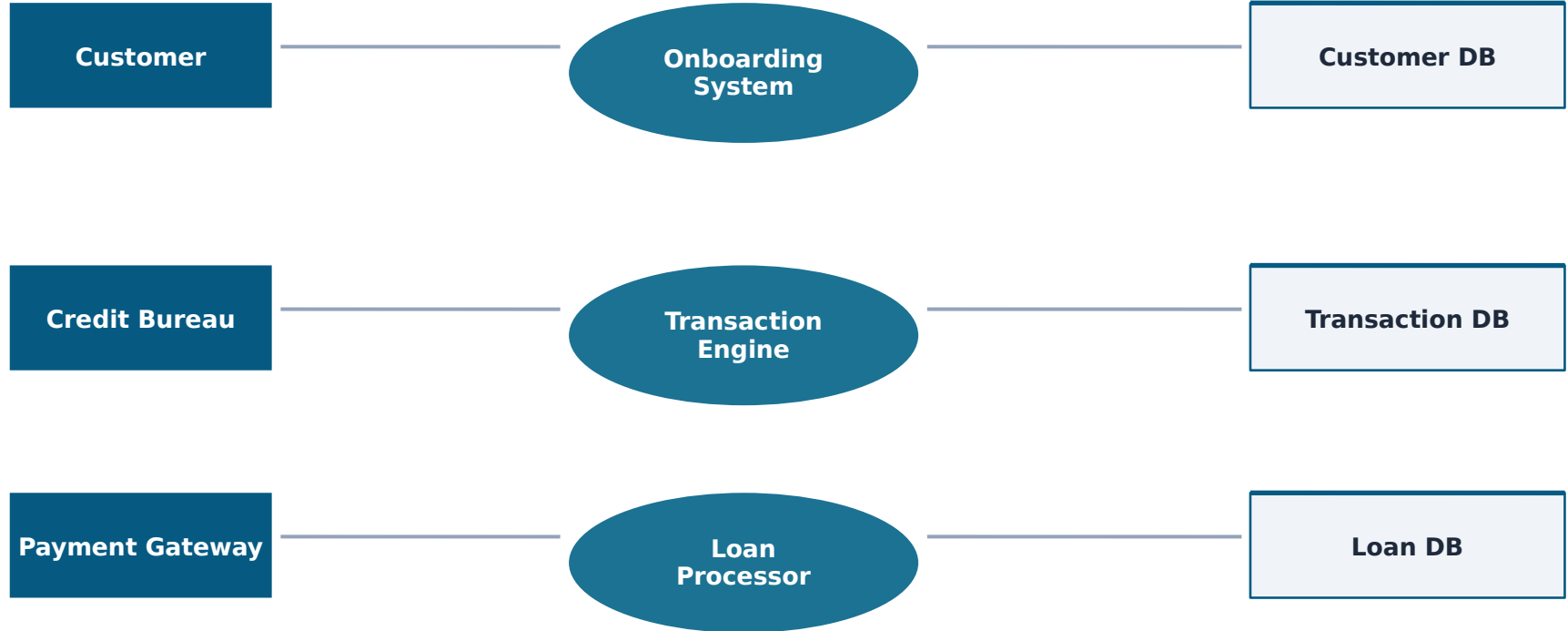


LEGEND

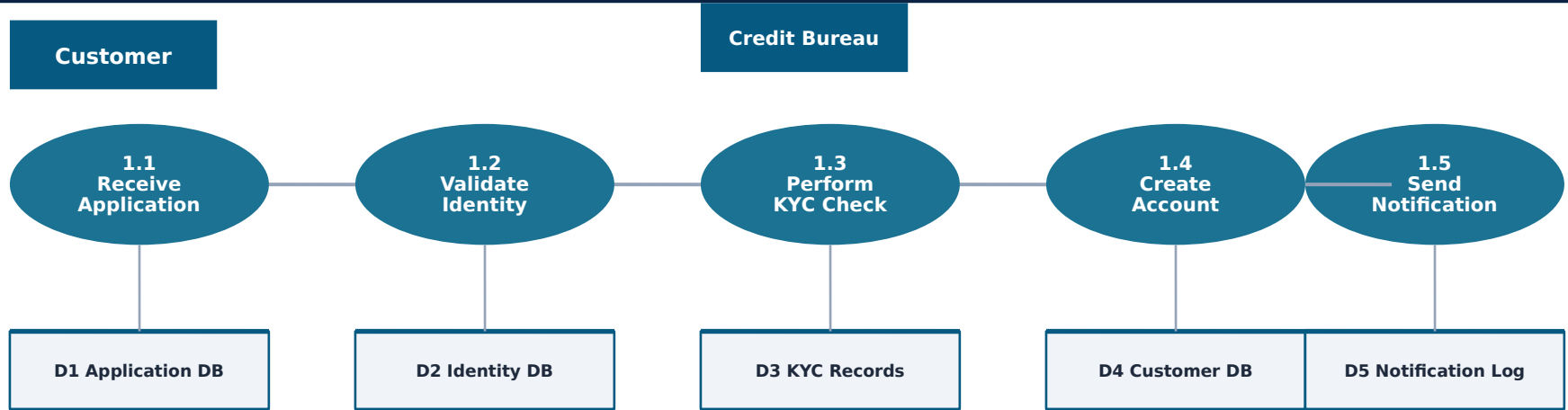
-  Start/End
-  Process Step
-  Decision

Key Components: Application Portal | Identity Verification | KYC Engine | Payment Gateway | Credit Bureau | Account Management | Notification Service | Loan Processor | Fraud Detection Engine

System Design: Data Flow Diagram (Level 0)

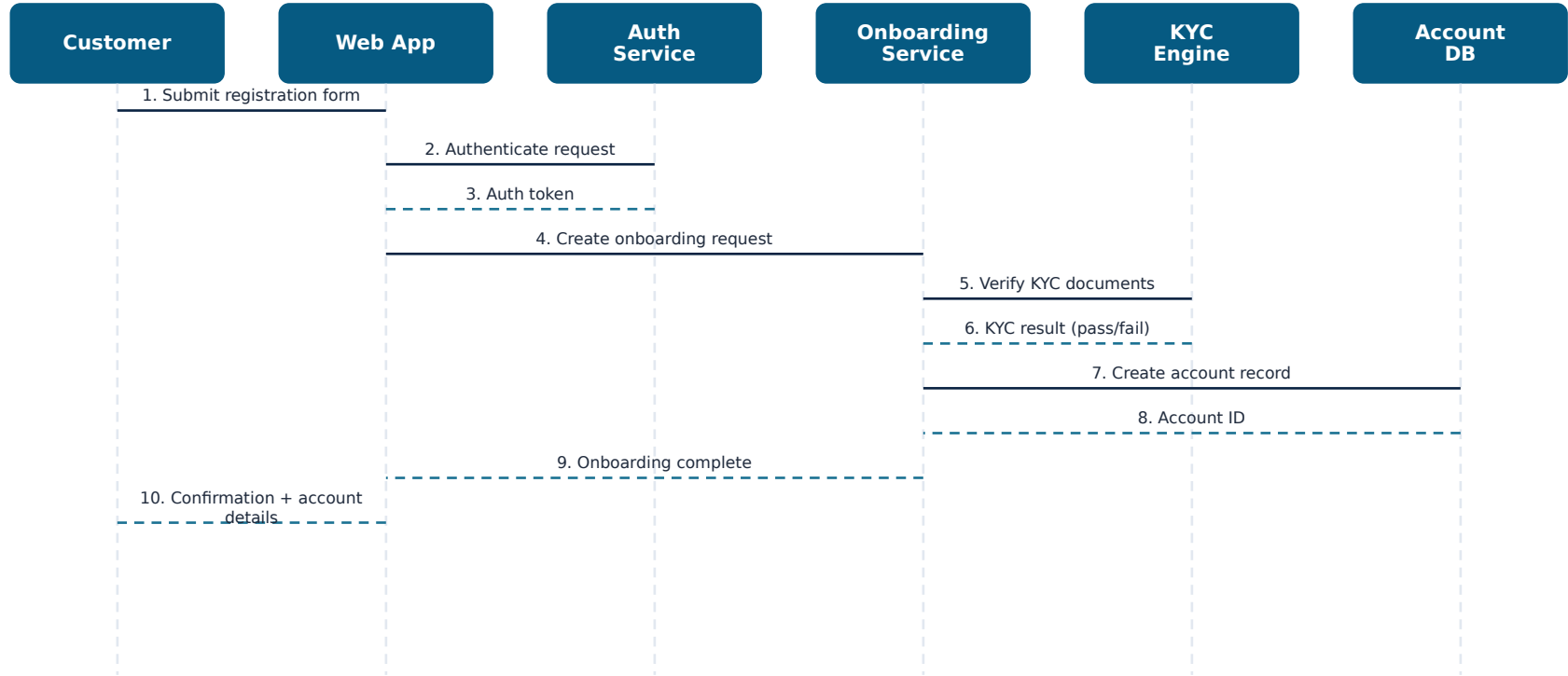


Data Flow Diagram (Level 1: Onboarding)



Data Flows: Application Data → 1.1 → Identity Info → 1.2 → KYC Request → 1.3 → Account Data → 1.4 → Confirmation → 1.5 → Email/SMS

Sequence Diagram: Customer Onboarding



System Architecture & Cloud Topology

Presentation Layer

Web App

Mobile App

API Gateway

Application Layer

Auth Service

Onboarding Svc

Transaction Svc

Loan Service

Data Layer

PostgreSQL

Redis Cache

Kafka MQ

IBM Cloud

Kubernetes Cluster

Container Registry

Load Balancer

WAF

Cloud Monitoring

Cloud Logging

Security Architecture Diagram

Perimeter Security

WAF

DDoS Protection

TLS 1.3 Termination

Network Security

VPC Isolation

Network ACLs

Firewall Rules

VPN Tunnels

Application Security

OAuth 2.0 / JWT

RBAC Authorization

Input Validation

API Rate Limiting

Data Security

AES-256 Encryption

Key Management

Data Masking

Audit Logging

COMPLIANCE

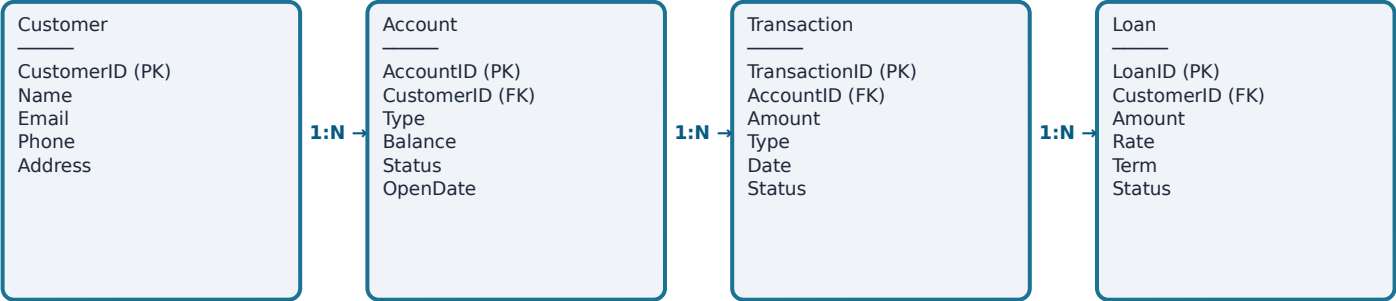
- PCI-DSS v4.0 compliant
- KYC/AML automated checks
- GDPR data handling
- SOC 2 Type II controls
- Quarterly pen testing
- SIEM integration
- Incident response SLA < 1hr
- Automated vulnerability scanning

Proof of Concept (PoC)

Section	Details
Objectives	Validate microservices architecture for the customer onboarding flow
Scope	Auth service + Onboarding service + KYC integration
Setup	Docker containers on IBM Cloud Kubernetes, PostgreSQL, REST APIs
Tools	Docker, Kubernetes, Jenkins, PostgreSQL, Kafka
Limitations	Onboarding flow only; no production data; single region deployment
Recommendations	Proceed with full build; add Redis caching; implement circuit breakers

Database Design: Conceptual & Logical ERD

CONCEPTUAL ERD



LOGICAL ERD — NORMALIZATION

- 1NF: All fields atomic
- 2NF: No partial dependencies
- 3NF: No transitive deps
- All tables in BCNF
- Referential integrity via FK

DATA DICTIONARY

Field	Type	Length	Nullabl e	Description
CustomerID	INT	11	No	Primary key, auto-increment
Name	VARCHAR	100	No	Customer full name
AccountID	INT	11	No	Primary key, auto-increment
Balance	DECIMAL	15,2	No	Current account balance
TransactionID	INT	11	No	Primary key, auto-increment

NORMALIZATION KEYS

Customer(CustomerID) → Account(CustomerID FK) → Transaction(AccountID FK) | Customer(CustomerID) → Loan(CustomerID FK)

DATA MIGRATION STRATEGY

- Phase 1: Extract legacy data from Oracle/DB2
- Phase 2: Transform and cleanse using ETL pipeline
- Phase 3: Load into PostgreSQL with validation
- Phase 4: Parallel run with data sync for 2 weeks
- Phase 5: Cutover with rollback plan ready
- Validation: Row counts, checksums, referential integrity checks post-migration

Test Plan & Test Cases (TDD/BDD)

Element	Details
Scope	All microservices, APIs, integration points, and database operations
Purpose	Validate functional and non-functional requirements against acceptance criteria
Objectives	100% critical path coverage; performance benchmarks met; zero P1 defects at launch
Deliverables	Test strategy document; test cases; test execution report; defect log; coverage report
Approach	TDD for unit tests; BDD for acceptance tests; automated regression suite
Environment	Staging cluster mirroring production configuration on IBM

TDD Example (Annotated)

```
# TC-001: Verify account creation returns ACTIVE status
# Priority: High | Linked Req: REQ-001
def test_account_creation():
    account = create_account(valid_data)
    assert account.status == "ACTIVE" # verify status
    assert account.id is not None     # verify ID assigned
    assert account.created_at is not None
```

BDD Scenario (Gherkin) — TC-002: Customer Onboarding | Priority: Critical | Linked Req: REQ-001, REQ-004

Feature: Customer Onboarding

Scenario: Successful account creation with valid KYC

Given a customer provides valid KYC documents including passport and proof of address

When they submit the onboarding form with all required fields

Then an account is created within 10 minutes with status ACTIVE

And a confirmation email is sent to the registered email address

And the account appears in the customer's dashboard

CI/CD Workflow & Monitoring Plan

CI WORKFLOW



CD PIPELINE



CONFIGURATION

- Jenkinsfile pipeline-as-code
- Helm charts for K8s deployment
- Docker Compose for local dev

BACKUP & RECOVERY

- Automated daily DB backups
- RPO: 1 hour; RTO: 4 hours
- Cross-region replication active

CLOUD COMPONENTS

- IBM Cloud Kubernetes Service
- IBM Container Registry
- IBM Cloud Object Storage

MONITORING FRAMEWORK



Key Findings

300%

Onboarding Time Overhead

Legacy systems increase onboarding time by 300% compared to industry benchmarks due to manual data entry and paper-based verification.

45 min

Incident Detection Delay

Lack of real-time monitoring means average incident detection takes 45 minutes, well above the 5-minute industry target.

23%

Resource Waste

Current monolithic architecture prevents independent scaling, causing 23% resource waste during off-peak hours.

Key Recommendations

1

Adopt Microservices Architecture

Enable independent deployment and scaling of each banking service for greater agility.

2

Implement Event-Driven Architecture

Use Kafka for real-time transaction processing and fraud detection across all channels.

3

Deploy CI/CD Pipeline

Automated testing and deployment to reduce release cycles from monthly to weekly.

4

Establish Centralized Monitoring

Prometheus and Grafana to achieve sub-5-minute incident detection and response.

Conclusion

- 1 The modernization initiative addresses critical operational inefficiencies in ABC Bank's digital infrastructure.
- 2 Cloud-native microservices architecture provides the scalability and resilience needed for modern banking.
- 3 Phased implementation minimizes risk while delivering incremental business value.
- 4 Success depends on stakeholder alignment, robust testing, and continuous monitoring post-deployment.

Appendix

REFERENCES

- IBM Cloud Documentation (2024)
- TOGAF Architecture Framework v10
- PCI-DSS Compliance Guide v4.0
- OWASP Security Testing Guide v4.2

TOOLS USED

- Draw.io for UML and process diagrams
- Jira for project tracking and RTM
- Jenkins for CI/CD pipeline automation
- Prometheus/Grafana for monitoring

SUPPORTING MATERIALS

- Detailed API specifications available in project repository
- Full stakeholder interview transcripts and meeting notes
- Complete test execution reports and defect logs